

Math 231A HW 9 (Final)

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Problem 1

Etingof Problem Sets 6.2:

Show that for $\mathfrak{g} = \mathfrak{gl}(n, \mathbb{C})$, the definition of a semisimple element (an element x such that $\text{ad } x$ is a semisimple operator) coincides to the usual definition of a semisimple operator.

Solution: The goal is to show that $x \in \mathfrak{g}$ (as a matrix in $\mathfrak{gl}(n, \mathbb{C}) := M_{n \times n}(\mathbb{C})$) has $\text{ad } x \in \text{End}(\mathfrak{g})$ being a semisimple operator $\iff x \in \text{End}(\mathbb{C}^n)$ is a semisimple operator. (Note: Here assume that $x \in \mathfrak{gl}(n, \mathbb{C})$ is a matrix under standard basis).

$\Leftarrow$: Suppose $x \in \text{End}(\mathbb{C}^n)$ is a semisimple operator (which over $\mathbb{C}$ it's equivalent to diagonalizability), then there exists basis $\alpha = \{v_1, \dots, v_n\} \subset \mathbb{C}^n$, such that the matrix of x under this basis is diagonal, which denotes as $D_x := \mathcal{M}(x, \alpha) = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}$ (where λ_i are not necessarily distinct).

Which, let the matrix $P := \begin{pmatrix} | & \dots & | \\ v_1 & \dots & v_n \\ | & \dots & | \end{pmatrix}$ be the matrix formed by $v_1, \dots, v_n$ (as column vectors under standard basis of $\mathbb{C}^n$), then as a matrix, one has $x = PD_xP^{-1}$ (since P is a change of basis matrix from basis α to standard basis).

Now, observe that each elementary matrix E_{ij} (with the i^{th} row j^{th} column entry being 1, and the rest being 0) satisfies the following:

$$\text{ad } D_x(E_{ij}) = D_x E_{ij} - E_{ij} D_x = \lambda_i E_{ij} - \lambda_j E_{ij} = (\lambda_i - \lambda_j) E_{ij} \quad (1.1)$$

Hence, the basis $\{E_{ij}\}_{1 \leq i, j \leq n} \subset \mathfrak{g}$ are also eigenvectors of $\text{ad } D_x$.

Then, we claim that the collection $\{PE_{ij}P^{-1}\}_{1 \leq i, j \leq n} \subset \mathfrak{g}$ is a basis such that the elements are eigenvectors of $\text{ad } x$:

- To show it's a basis, it suffices to show it's linearly independent (since $\dim \mathfrak{g} = n^2$, and the list has length n^2): If $\sum_{i,j} a_{ij} PE_{ij}P^{-1} = P \left(\sum_{i,j} a_{ij} E_{ij} \right) P^{-1} = 0$, one has $\sum_{i,j} a_{ij} E_{ij} = 0$, hence each $a_{ij} = 0$ (by linear independence of $\{E_{ij}\}_{1 \leq i, j \leq n}$). So, the list is linearly independent, hence a basis.
- Each of them is an eigenvector of $\text{ad } x$, because they satisfy the following:

$$\begin{aligned}
\operatorname{ad} x(PE_{ij}P^{-1}) &= x(PE_{ij}P^{-1}) - (PE_{ij}P^{-1})x \\
&= (PD_xP^{-1})(PE_{ij}P^{-1}) - (PE_{ij}P^{-1})(PD_xP^{-1}) \\
&= P(D_xE_{ij} - E_{ij}D_x)P^{-1} \\
&= P(\operatorname{ad} D_x(E_{ij}))P^{-1} \\
&= (\lambda_i - \lambda_j)PE_{ij}P^{-1}
\end{aligned} \tag{1.2}$$

So, $\{PE_{ij}P^{-1}\}_{1 \leq i, j \leq n} \subset \mathfrak{g}$ is indeed a basis, consists of eigenvectors of $\operatorname{ad} x$, showing $\operatorname{ad} x \in \operatorname{End}(\mathfrak{g})$ is a semisimple operator, hence $x \in \mathfrak{g}$ is a semisimple element.

$\Rightarrow$: Suppose $\operatorname{ad} x \in \operatorname{End}(\mathfrak{g})$ is a semisimple operator. Let $x = x_s + x_n$ (where $x_s, x_n \in \mathfrak{g} = \operatorname{End}(\mathbb{C}^n)$ are semisimple and nilpotent operators on $\mathbb{C}^n$ respectively) such that $x_s x_n = x_n x_s$, or $[x_s, x_n] = 0$ (which can be done using Jordan decomposition). In particular, $x x_s = x_s x$, showing the following:

$$\operatorname{ad} x(\operatorname{ad} x_s(A)) = [x, [x_s, A]] = -[x_s, [A, x]] - [A, [x, x_s]] = [x_s, [x, A]] = \operatorname{ad} x_s(\operatorname{ad} x(A)) \tag{1.3}$$

This shows that $\operatorname{ad} x, \operatorname{ad} x_s$ are commuting operators. Which, based on the converse above, since $\operatorname{ad} x_s$ is also semisimple, it is diagonalizable; with $\operatorname{ad} x_s, \operatorname{ad} x$ being two diagonalizable operators that commute, they're simultaneously diagonalizable.

Let $A_1, \dots, A_{n^2} \in \mathfrak{g}$ be the basis consists of common eigenvectors of $\operatorname{ad} x$ and $\operatorname{ad} x_s$ (with eigenvalues $\lambda_1, \dots, \lambda_{n^2}$ and $\mu_1, \dots, \mu_{n^2}$ respectively). Then, one has the following:

$$\lambda_i A_i = \operatorname{ad} x(A_i) = \operatorname{ad}(x_s + x_n)(A_i) = \operatorname{ad} x_s(A_i) + \operatorname{ad} x_n(A_i) = \mu_i A_i + \operatorname{ad} x_n(A_i) \tag{1.4}$$

Hence, $\operatorname{ad} x_n(A_i) = (\lambda_i - \mu_i)A_i$, showing that $\operatorname{ad} x_n$ is also a diagonalizable operator; however, notice that since $x_n \in \operatorname{End}(\mathbb{C}^n)$ is nilpotent, hence so is $\operatorname{ad} x_n$. Then, being semisimple and nilpotent implies $\operatorname{ad} x_n = 0$, so $x_n \in \mathfrak{z}(\mathfrak{g})$, the center of $\mathfrak{gl}(n, \mathbb{C})$.

Finally, since the center of $\mathfrak{gl}(n, \mathbb{C})$ are all the complex scalar multiples of identity, the $x_n = \lambda \cdot I$ for some $\lambda \in \mathbb{C}$; then with nilpotency of x_n as an operator on $\mathbb{C}^n$, one must have $\lambda = 0$. So, $x = x_s$ is a semisimple operator on $\mathbb{C}^n$.

The above two directions shows the equivalency of $x \in \mathfrak{gl}(n, \mathbb{C})$ being a semisimple operator on $\mathbb{C}^n$, and $\operatorname{ad} x \in \operatorname{End}(\mathfrak{g})$ being a semisimple operator.

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Problem 2

Etingof Problem Sets 6.3:

Show that if $\mathfrak{h} \subset \mathfrak{g}$ is a Cartan subalgebra in a complex semisimple Lie algebra, then $\mathfrak{h}$ is a nilpotent subalgebra which coincides with its normalizer $n(\mathfrak{h}) = \{x \in \mathfrak{g} \mid \text{ad } x \cdot \mathfrak{h} \subset \mathfrak{h}\}$. (This is the usual definition of a Cartan subalgebra which can be used for any Lie algebra, not necessarily a semisimple one).

Solution: First, based on Etingof's Lecture Notes, Cartan Subalgebra is defined to be a Toral subalgebra $\mathfrak{h}$ (a subalgebra consists of semisimple elements) such that $\mathfrak{g}_0 = \mathfrak{h}$. So, since $\mathfrak{h}$ is a toral subalgebra (in particular, it's abelian), it is nilpotent (since $[\mathfrak{h}, \mathfrak{h}] = 0$).

It's clear that $\mathfrak{h} \subseteq n(\mathfrak{h})$, since for all $x, y \in \mathfrak{h}$, one has $\text{ad } x(y) = [x, y] = 0 \in \mathfrak{h}$, showing $\text{ad } x \cdot \mathfrak{h} \subset \mathfrak{h}$, or $x \in n(\mathfrak{h})$.

Now, given any $x \in n(\mathfrak{h})$ and $y \in \mathfrak{h}$, one has $\text{ad } x(y) = [x, y] \in \mathfrak{h} = \mathfrak{g}_0$, so this implies that $\text{ad } y([x, y]) = -(\text{ad } y)^2(x) = 0$. But, since y is semisimple / diagonalizable (since $\mathfrak{h}$ is a Toral subalgebra, and over $\mathbb{C}$ semisimple and diagonalizable are equivalent), $(\text{ad } y)^2(x) = 0$ implies $\text{ad } y(x) = 0$ (since any diagonalizable operator T has a stabilized kernel, i.e. $\ker(T) = \ker(T^n)$ for all integer $n > 0$), hence $x \in \mathfrak{g}_0 = \mathfrak{h}$ (since $\text{ad } y(x) = 0$ for all $y \in \mathfrak{h}$). So, one concludes that $n(\mathfrak{h}) \subseteq \mathfrak{h}$.

As a result, $n(\mathfrak{h}) = \mathfrak{h}$, showing a Cartan Subalgebra of a complex semisimple Lie algebra, is both Nilpotent, and being precisely its own normalizer.

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Problem 3

Etingof Problem Sets 6.4:

Let $\mathfrak{g}$ be a complex Lie algebra which has a root decomposition:

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_{\alpha} \quad (3.1)$$

where R is a finite subset in $\mathfrak{h}^* \setminus \{0\}$, $\mathfrak{h}$ is commutative and for $h \in \mathfrak{h}$, $x \in \mathfrak{g}_{\alpha}$, we have $[h, x] = \langle h, \alpha \rangle x$. Show that then $\mathfrak{g}$ is semisimple, and $\mathfrak{h}$ is a Cartan subalgebra.

(Correction: For any $\alpha \in \mathfrak{h}^*$, the Killing form K also gives a non-degenerate pairing on $\mathfrak{g}_{\alpha} \times \mathfrak{g}_{-\alpha}$, which implies $\alpha \in R \iff -\alpha \in R$, while the restriction of K to $\mathfrak{h}$ is non-degenerate).

Solution: First, given that the Killing form K restricts to a non-degenerate pairing $\mathfrak{g}_{\alpha} \times \mathfrak{g}_{-\alpha}$, it's clear that the Killing form is non-degenerate, since for any $x \in \mathfrak{h}$, one has some $y \in \mathfrak{h}$ such that $K(x, y) \neq 0$, while for any $\alpha \in R$, since $-\alpha \in R$, then any $x' \in \mathfrak{g}_{\alpha}$ has some nonzero $y' \in \mathfrak{g}_{-\alpha}$, such that $K(x', y') \neq 0$. Since $\mathfrak{g} = \mathfrak{h} \oplus \left(\bigoplus_{\alpha \in R} \mathfrak{g}_{\alpha} \right)$, then choose a basis according to this direct sum, one has the Killing Form being non-degenerate for all the basis elements, showing $\ker(K) = 0$. Hence, as a result of Cartan Semisimplicity Theorem, $\mathfrak{g}$ is semisimple.

Now, to show that $\mathfrak{h}$ is a Cartan subalgebra, first it's clear that $\mathfrak{h}$ is Toral, since it's first abelian, and for any $x \in \mathfrak{h}$, choose a basis according to the Root Decomposition $\mathfrak{g} = \mathfrak{h} \oplus \left(\bigoplus_{\alpha \in R} \mathfrak{g}_{\alpha} \right)$ (i.e. a basis consists of elements in each direct sum), then one has $\text{ad } x$ acts diagonally on this basis (since each $\mathfrak{g}_{\alpha}$ is an eigenspace of $\text{ad } x$, with weight $\alpha(x)$), while any $y \in \mathfrak{h}$ has $\text{ad } x(y) = 0$ by abelian

property of $\mathfrak{h}$), showing $\text{ad } x$ is diagonalizable, hence semisimple. So, any $x \in \mathfrak{h}$ is a semisimple element, showing $\mathfrak{h}$ is toral.

Finally, since for any $x, y \in \mathfrak{h}$, the abelian property guarantees $\text{ad } x(y) = 0$, showing $y \in \mathfrak{g}_0$ (since $\text{ad } x(y) = 0 \cdot y$ for all $x \in \mathfrak{h}$), so $\mathfrak{h} \subseteq \mathfrak{g}_0$. Also, for any $y \in \mathfrak{g}_0$ (say $y = x_0 + x_1 + \dots + x_n$, with $x_0 \in \mathfrak{h}$, and each $x_i \in \mathfrak{g}_{\alpha_i}$ based on the root decomposition of $\mathfrak{g}$), since for any $\alpha_i \in R$, there exists $z_i \in \mathfrak{h}$ such that $\alpha_i(z_i) \neq 0$, then one has the following:

$$0 = \text{ad } z_i(y) = [z_i, x] + [z_i, x_1] + \dots + [z_i, x_n] = \alpha_1(z_i)x_1 + \dots + \alpha_i(z_i)x_i + \dots + \alpha_n(z_i)x_n \quad (3.2)$$

Which, each $\alpha_j(z_i)x_j \in \mathfrak{g}_{\alpha_j}$, so by the direct sum property this enforces each $\alpha_j(z_i)x_j = 0$. But, with $\alpha_i(z_i) \neq 0$, one must have $x_i = 0$ for this to be true. Hence, with $i \in \{1, \dots, n\}$ being arbitrary, one must have each $x_i = 0$, showing $y = x_0 + x_1 + \dots + x_n = x_0 \in \mathfrak{h}$. So, $\mathfrak{g}_0 \subseteq \mathfrak{h}$, showing $\mathfrak{g}_0 = \mathfrak{h}$. Hence, $\mathfrak{h}$ is a Cartan subalgebra.

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Problem 4

Etingof Problem Sets 6.5:

Let $\mathfrak{h} \subset \mathfrak{so}(4, \mathbb{C})$ be the subalgebra consisting of matrices of the form

$$\begin{pmatrix} 0 & a & 0 & 0 \\ -a & 0 & 0 & 0 \\ 0 & 0 & 0 & b \\ 0 & 0 & -b & 0 \end{pmatrix} \quad (4.1)$$

Show that then $\mathfrak{h}$ is a Cartan subalgebra and find the corresponding root decomposition.

Solution: First, define the following elements as basis of $\mathfrak{so}(4, \mathbb{C})$:

$$\begin{aligned} J_1 &:= \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, & J_2 &:= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, & J_3 &:= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \\ J_4 &:= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, & J_5 &:= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, & J_6 &:= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \end{aligned} \quad (4.2)$$

Which, $\mathfrak{h} := \text{span}\{J_1, J_6\}$.

I. $\mathfrak{h}$ is Abelian:

First, $\mathfrak{h}$ is a subalgebra, in particular because it is abelian due to the following relation:

$$[J_1, J_6] = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = 0 - 0 = 0 \quad (4.3)$$

Hence, given any $a, b, c, d \in \mathbb{C}$, one has $[aJ_1 + bJ_6, cJ_1 + dJ_6] = ac[J_1, J_1] + (ad - bc)[J_1, J_6] + bd[J_6, J_6] = 0$, showing $\mathfrak{h}$ is abelian.

II. $\mathfrak{h}$ is Toral:

To show that $\mathfrak{h}$ is toral, we'll show all elements in here are semisimple, in particular we'll show that J_1, J_6 are semisimple by finding their eigenvalues / eigenvectors.

Given any $a, b \in \mathbb{C}$, consider the following commutation relations of $aJ_1 + bJ_6$ with the remaining basis elements:

$$\begin{aligned} [aJ_1 + bJ_6, J_2] &= -bJ_3 - aJ_4, & [aJ_1 + bJ_6, J_3] &= bJ_2 - aJ_5 \\ [aJ_1 + bJ_6, J_4] &= aJ_2 - bJ_5, & [aJ_1 + bJ_6, J_5] &= aJ_3 + bJ_4 \end{aligned} \quad (4.4)$$

Which, given any $\sum_{i=1}^6 c_i J_i$, one has the following:

- Suppose it's an eigenvector of $\text{ad } J_1$ with eigenvalue λ (which $a = 1, b = 0$), then one has the following:

$$\sum_{i=1}^6 \lambda \cdot c_i J_i = \left[J_1, \sum_{i=1}^6 c_i J_i \right] = c_2 \cdot (-J_4) + c_3 \cdot (-J_5) + c_4 \cdot J_2 + c_5 \cdot J_3 \quad (4.5)$$

Hence, one has $\lambda \cdot c_1 = \lambda \cdot c_6 = 0$, $\lambda \cdot c_2 = c_4$, $\lambda \cdot c_3 = c_5$, $\lambda \cdot c_4 = -c_2$, and $\lambda \cdot c_5 = -c_3$

- The first relation $\lambda \cdot c_1 = \lambda \cdot c_6 = 0$ has one possible solution $\lambda = 0$. Which, this enforces $c_2 = c_3 = c_4 = c_5 = 0$, showing $\sum_{i=1}^6 c_i J_i = c_1 J_1 + c_6 J_6 \in \mathfrak{h}$ (this is also an equivalent statement, where $\lambda = 0$ iff $\sum_{i=1}^6 c_i J_i \in \mathfrak{h}$, because of the coefficient).
- Now, suppose $\sum_{i=1}^6 c_i J_i \notin \mathfrak{h}$, which implies one of the $c_2, c_3, c_4, c_5 \neq 0$. For definiteness say $c_2 \neq 0$, then one has $c_4 = \lambda c_2$ and $-c_2 = \lambda c_4$, showing $-c_2 = \lambda^2 c_2$, or $(\lambda^2 + 1)c_2 = 0$, hence $\lambda^2 = -1$, showing $\lambda = \pm i$. Which, they provide the following relations:

$$\begin{aligned} \text{(a) } \lambda = i &\implies (c_2, c_3, c_4, c_5) = (c_2, c_3, ic_2, ic_3) \\ \text{(b) } \lambda = -i &\implies (c_2, c_3, c_4, c_5) = (c_2, c_3, -ic_2, -ic_3) \end{aligned} \quad (4.6)$$

- Suppose it's an eigenvector of $\text{ad } J_6$ with eigenvalue μ (which $a = 0, b = 1$), then one has the following:

$$\sum_{i=1}^6 \mu \cdot c_i J_i = \left[J_6, \sum_{i=1}^6 c_i J_i \right] = c_2 \cdot (-J_3) + c_3 \cdot J_2 + c_4 \cdot (-J_5) + c_5 \cdot J_4 \quad (4.7)$$

Hence, one has $\mu \cdot c_1 = \mu \cdot c_6 = 0$, $\mu \cdot c_2 = c_3$, $\mu \cdot c_3 = -c_2$, $\mu \cdot c_4 = c_5$, and $\mu \cdot c_5 = -c_4$.

Again if $\mu = 0$ is equivalent to $\sum_{i=1}^6 c_i J_i = c_1 J_1 + c_6 J_6 \in \mathfrak{h}$; else if $\sum_{i=1}^6 c_i J_i \notin \mathfrak{h}$ (i.e. one of the c_2, c_3, c_4, c_5 is nonzero), one can prove again that $\mu = \pm i$. Which, they provide the following relations:

$$\begin{aligned} \text{(c) } \mu = i &\implies (c_2, c_3, c_4, c_5) = (c_2, ic_3, c_4, ic_5) \\ \text{(d) } \mu = -i &\implies (c_2, c_3, c_4, c_5) = (c_2, -ic_3, c_4, -ic_5) \end{aligned} \quad (4.8)$$

Which, combining the relations (a),(b),(c),(d), one gets:

$$\begin{aligned}
(a), (c) &\implies (c_2, c_3, c_4, c_5) = (c_2, ic_2, ic_2, -c_2) \\
(a), (d) &\implies (c_2, c_3, c_4, c_5) = (c_2, -ic_2, ic_2, c_2) \\
(b), (c) &\implies (c_2, c_3, c_4, c_5) = (c_2, ic_2, -ic_2, c_2) \\
(b), (d) &\implies (c_2, c_3, c_4, c_5) = (c_2, -ic_2, -ic_2, -c_2)
\end{aligned} \tag{4.9}$$

Which, these four collection of entries in fact form a linearly independent set in $\mathbb{C}^4$, which mapping over to $\mathfrak{so}(4, \mathbb{C})$ by corresponding the entries to J_2, J_3, J_4, J_5 respectively, the above four elements each are eigenvectors of $\text{ad } J_1$ and $\text{ad } J_6$, hence are eigenvectors of $\text{ad } (aJ_1 + bJ_6)$ for any $aJ_1 + bJ_6 \in \mathfrak{h}$; together with $\text{ad } x \Big|_{\mathfrak{h}} = 0$ for any $x \in \mathfrak{h}$, this shows that $\text{ad } x$ has eigenvectors that span the whole $\mathfrak{so}(4, \mathbb{C})$ (for all $x \in \mathfrak{h}$), showing all $x \in \mathfrak{h}$ are semisimple elements, hence $\mathfrak{h}$ is toral.

III. $\mathfrak{h}$ is a Cartan Subalgebra:

From **II**, it's clear that if $\sum_{i=1}^6 c_i J_i$ is a common zero eigenvector of elements in $\mathfrak{h}$, implies $\sum_{i=1}^6 c_i J_i = c_1 J_1 + c_6 J_6 \in \mathfrak{h}$ (since any zero eigenvector of $\text{ad } J_1$ and $\text{ad } J_6$ has proven to be in $\mathfrak{h}$), so one has $\mathfrak{g}_0 \subseteq \mathfrak{h}$ (since for any $y \in \mathfrak{g}_0$, any $x \in \mathfrak{h}$ has $\text{ad } x(y) = 0$, choose $x = J_1, J_6$ implies $y \in \mathfrak{h}$); also, since $\mathfrak{h}$ is abelian (proven in **I**), then any $x, y \in \mathfrak{h}$ has $\text{ad } x(y) = 0$, showing $y \in \mathfrak{g}_0$, or $\mathfrak{h} \subseteq \mathfrak{g}_0$.

Hence, $\mathfrak{g}_0 = \mathfrak{h}$, showing $\mathfrak{h}$ is a Cartan Subalgebra (based on the definition in Etingof's Lecture Notes).

IV. Root Decomposition of $\mathfrak{so}(4, \mathbb{C})$ by $\mathfrak{h}$:

Using the relations in **II**, let $\lambda = \pm i$, $\mu = \pm i$ be the eigenvalues of $\text{ad } J_1, \text{ad } J_6$ respectively, then their common eigenvectors are of the following form:

$$\begin{aligned}
(a), (c) &\implies (c_2, c_3, c_4, c_5) = (c_2, ic_2, ic_2, -c_2), \quad v_{i,i} = c_2(J_2 + iJ_3 + iJ_4 - J_5) \\
(a), (d) &\implies (c_2, c_3, c_4, c_5) = (c_2, -ic_2, ic_2, c_2), \quad v_{i,-i} = c_2(J_2 - iJ_3 + iJ_4 + J_5) \\
(b), (c) &\implies (c_2, c_3, c_4, c_5) = (c_2, ic_2, -ic_2, c_2), \quad v_{-i,i} = c_2(J_2 + iJ_3 - iJ_4 + J_5) \\
(b), (d) &\implies (c_2, c_3, c_4, c_5) = (c_2, -ic_2, -ic_2, -c_2), \quad v_{-i,-i} = c_2(J_2 - iJ_3 - iJ_4 + J_5)
\end{aligned} \tag{4.10}$$

Where $v_{\lambda,\mu}$ denotes the eigenvector of $\text{ad } J_1, \text{ad } J_6$ with eigenvalue λ, μ respectively (where $\lambda, \mu = \pm i$). Which, for any $a, b \in \mathbb{C}$, this implies the following:

$$\text{ad } (aJ_1 + bJ_6) \cdot v_{\lambda,\mu} = a \cdot \lambda v_{\lambda,\mu} + b \cdot \mu v_{\lambda,\mu} = (a\lambda + b\mu)v_{\lambda,\mu} \tag{4.11}$$

Hence, each $v_{\lambda,\mu}$ is a common eigenvector for all $x = aJ_1 + bJ_6 \in \mathfrak{h}$, with eigenvalue $a\lambda + b\mu$. Then, define linear functionals $\alpha_{\lambda,\mu} \in \mathfrak{h}^*$ as follow for $\lambda, \mu = \pm i$:

$$\alpha_{\lambda,\mu}(aJ_1 + bJ_6) := a\lambda + b\mu \tag{4.12}$$

Then, the above relations shows that $\mathbb{C}v_{\lambda,\mu} = \mathfrak{g}_{\alpha_{\lambda,\mu}}$. Hence, with each $v_{\lambda,\mu}$ being linearly independent (and together with $J_1, J_6 \in \mathfrak{h}$, one can see $J_1, J_6, v_{i,i}, v_{i,-i}, v_{-i,i}, v_{-i,-i}$ is a linearly independent list, hence a basis for $\mathfrak{so}(4, \mathbb{C})$), then one has the following root decomposition of $\mathfrak{so}(4, \mathbb{C})$:

$$\mathfrak{so}(4, \mathbb{C}) = (\mathbb{C}J_1 \oplus \mathbb{C}J_6) \oplus \left(\bigoplus_{\lambda,\mu=\pm i} \mathbb{C}v_{\lambda,\mu} \right) = \mathfrak{h} \oplus \left(\bigoplus_{\lambda,\mu=\pm i} \mathfrak{g}_{\alpha_{\lambda,\mu}} \right) \tag{4.13}$$

Where, a root system for the above decomposition is $R = \{\alpha_{\lambda,\mu} \mid \lambda, \mu = \pm i\}$, and each $\alpha_{\lambda,\mu} \in \mathfrak{h}^*$ satisfies $\alpha_{\lambda,\mu}(aJ_1 + bJ_6) = a\lambda + b\mu$.

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Problem 5

- (1) Define a bilinear form B on $W = \Lambda^2 \mathbb{C}^4$ by $w_1 \wedge w_2 = B(w_1, w_2)e_1 \wedge e_2 \wedge e_3 \wedge e_4$. Show that B is a symmetric non-degenerate form and construct an orthonormal basis for B .
- (2) Let $\mathfrak{g} = \mathfrak{so}(W, B) = \{x \in \mathfrak{gl}(W) \mid B(xw_1, w_2) + B(w_1, xw_2) = 0\}$. Show that $\mathfrak{g} \cong \mathfrak{so}(6, \mathbb{C})$.
- (3) Show that the form B is invariant under the natural action of $\mathfrak{sl}(4, \mathbb{C})$ on $\Lambda^2 \mathbb{C}^4$.
- (4) Using results of the previous parts, construct a homomorphism $\mathfrak{sl}(4, \mathbb{C}) \rightarrow \mathfrak{so}(6, \mathbb{C})$ and prove that it is an isomorphism.

Solution:

- (1) First, recall that $\Lambda^2 \mathbb{C}^4$ has basis $\{e_1 \wedge e_2, e_1 \wedge e_3, e_1 \wedge e_4, e_2 \wedge e_3, e_2 \wedge e_4, e_3 \wedge e_4\}$. Which, given $w_1 = \sum_{i < j} c_{ij} e_i \wedge e_j, w_2 = \sum_{k < l} d_{kl} e_k \wedge e_l \in \Lambda^2 \mathbb{C}^4$, one has the following relation:

$$w_1 \wedge w_2 = (c_{12}d_{34} - c_{13}d_{24} + c_{14}d_{23} + c_{23}d_{14} - c_{24}d_{13} + c_{34}d_{12})e_1 \wedge e_2 \wedge e_3 \wedge e_4 \quad (5.1)$$

So, one has $B(w_1, w_2) = (c_{12}d_{34} - c_{13}d_{24} + c_{14}d_{23} + c_{23}d_{14} - c_{24}d_{13} + c_{34}d_{12})$. Which, this form is symmetric, since $B(w_2, w_1)$ is done by swapping the coefficients c and d , which provides the following:

$$B(w_2, w_1) = (d_{12}c_{34} - d_{13}c_{24} + d_{14}c_{23} + d_{23}c_{14} - d_{24}c_{13} + d_{34}c_{12}) = B(w_1, w_2) \quad (5.2)$$

This shows that B is symmetric.

Then, to show it's non-degenerate, it suffices to check the non-degeneracy for the provided basis:

$$B(e_1 \wedge e_2, e_3 \wedge e_4) = 1, \quad B(e_1 \wedge e_3, e_2 \wedge e_4) = -1, \quad B(e_1 \wedge e_4, e_2 \wedge e_3) = 1 \quad (5.3)$$

Which, by symmetric property of B , the above shows the non-degeneracy on all the basis elements, showing B is non-degenerate.

Finally, notice that $B(e_i \wedge e_j, e_k \wedge e_l) = 0$ if $\{i, j\} \cap \{k, l\} \neq \emptyset$ (since if one of the basis elements appear more than once, the wedge product is 0), then here is a proposed orthonormal basis based on such observation and the previous equations:

$$\left\{ \begin{aligned} f_1 &= \frac{1}{\sqrt{2}}(e_1 \wedge e_2 + e_3 \wedge e_4), f_2 = \frac{i}{\sqrt{2}}(e_1 \wedge e_2 - e_3 \wedge e_4), f_3 = \frac{i}{\sqrt{2}}(e_1 \wedge e_3 + e_2 \wedge e_4), \\ f_4 &= \frac{1}{\sqrt{2}}(e_1 \wedge e_3 - e_2 \wedge e_4), f_5 = \frac{1}{\sqrt{2}}(e_1 \wedge e_4 + e_2 \wedge e_3), f_6 = \frac{i}{\sqrt{2}}(e_1 \wedge e_4 - e_2 \wedge e_3) \end{aligned} \right\} \quad (5.4)$$

Which, they satisfied the following equations for normalizability (for simplicity, only the case involving $e_1 \wedge e_2, e_3 \wedge e_4$ will be demonstrated, since the rest is the same concept):

$$\begin{aligned} \frac{1}{2}B(e_1 \wedge e_2 + e_3 \wedge e_4, e_1 \wedge e_2 + e_3 \wedge e_4) &= \frac{1}{2} \cdot 2B(e_1 \wedge e_2, e_3 \wedge e_4) = 1 \\ -\frac{1}{2}B(e_1 \wedge e_2 - e_3 \wedge e_4, e_1 \wedge e_2 - e_3 \wedge e_4) &= -\frac{1}{2} \cdot (-2)B(e_1 \wedge e_2, e_3 \wedge e_4) = 1 \end{aligned} \quad (5.5)$$

Also, for orthogonality, it suffices to check the case between the two elements involving the same standard basis elements (for other ones, for instance between $x = \frac{1}{\sqrt{2}}(e_1 \wedge e_2 + e_3 \wedge e_4)$

e_4) and $y = \frac{1}{\sqrt{2}}(e_1 \wedge e_3 - e_2 \wedge e_4)$, notice that between any two standard basis elements involved in x and y , the indices have nontrivial intersection, like in between $e_1 \wedge e_2$ involved in x and $e_1 \wedge e_3$ involved in y , they shared e_1 . So, one has $B(x, y) = 0$. Which, for definiteness, using the ones involving $e_1 \wedge e_2$ and $e_3 \wedge e_4$, one has:

$$\begin{aligned} \frac{i}{2}B(e_1 \wedge e_2 + e_3 \wedge e_4, e_1 \wedge e_2 - e_3 \wedge e_4) &= \frac{i}{2}(B(e_3 \wedge e_4, e_1 \wedge e_2) - B(e_1 \wedge e_2, e_3 \wedge e_4)) \\ &= 0 \end{aligned} \quad (5.6)$$

Which, the proposed list is indeed orthonormal under bilinear form B . Which, with the list having a length of $6 = \dim(\Lambda^2 \mathbb{C}^4)$, it's an orthonormal basis under bilinear form B .

- (2) Using the orthonormal basis $\{f_1, \dots, f_6\}$ (with respect to B) proposed in (1), if consider $\text{End}(W) \cong M_{6 \times 6}(\mathbb{C}) = \mathfrak{gl}(6, \mathbb{C})$ under this basis, then for any operator $x \in \mathfrak{g} \subseteq \text{End}(W)$, let $\mathcal{M}(x) = (a_{ij})_{1 \leq i, j \leq 6}$ be the matrix, one has the following based on orthonormality:

$$\sum_{i=1}^6 a_{ij} f_i = x(f_j) = \sum_{i=1}^6 B(x(f_j), f_i) f_i \implies a_{ij} = B(x(f_j), f_i) \quad (5.7)$$

Hence, with the characterization that $B(xw_1, w_2) + B(w_1, xw_2) = 0$ (or $B(xw_1, w_2) = -B(xw_2, w_1)$), one has $a_{ij} = B(xf_j, f_i) = -B(xf_i, f_j) = -a_{ji}$.

Hence, as a matrix one has $\mathcal{M}(x) \in \mathfrak{so}(6, \mathbb{C})$. So, since the matrix representative under this basis one has $\text{End}(W) \cong M_{n \times n}(\mathbb{C}) = \mathfrak{gl}(6, \mathbb{C})$, with each $x \in \mathfrak{g}$ corresponds to a matrix $\mathcal{M}(x) = (a_{ij})$ such that $a_{ij} = -a_{ji}$ (which is a skew-symmetric matrix), one has $\mathcal{M}(x) \in \mathfrak{so}(6, \mathbb{C})$, showing under the matrix map isomorphism $\mathcal{M} : \text{End}(W) \xrightarrow{\sim} \mathfrak{gl}(n, \mathbb{C})$, one has $\mathcal{M}(\mathfrak{g}) \subseteq \mathfrak{so}(6, \mathbb{C})$.

Conversely, given any matrix $A = (a_{ij}) \in \mathfrak{so}(6, \mathbb{C})$, let $x := \mathcal{M}^{-1}(A) \in \text{End}(W)$, then its action on the basis element satisfies $a_{ij} = B(x(f_j), f_i)$ (which is true for general operators in $\text{End}(W)$), then by skew-symmetric property of A , $B(x(f_j), f_i) = a_{ij} = -a_{ji} = -B(x(f_i), f_j)$, or $B(x(f_j), f_i) + B(x(f_i), f_j) = 0$. With this action being true on all pairs of basis, element, one has $B(xw_1, w_2) + B(w_1, xw_2) = 0$ for all $w_1, w_2 \in W$. So, $x \in \mathfrak{g}$.

This shows that $\mathcal{M}^{-1}(\mathfrak{so}(6, \mathbb{C})) \subseteq \mathfrak{g}$, or $\mathfrak{so}(6, \mathbb{C}) \subseteq \mathcal{M}(\mathfrak{g})$, finally showing that $\mathfrak{so}(6, \mathbb{C}) = \mathcal{M}(\mathfrak{g})$. With, $\mathcal{M}$ being an isomorphism, $\mathfrak{g} \cong \mathfrak{so}(6, \mathbb{C})$.

- (3) First, recall the fact that $B(w \wedge x, y \wedge z) = 0$, if $\{w, x\} \cap \{y, z\} \neq \emptyset$; also, recall that for any $x \in \mathfrak{sl}(4, \mathbb{C})$ (where under the standard basis, let the matrix $x = (a_{ij})$, so $x(e_j) = \sum_{i=1}^4 a_{ij} e_i$), the action on the tensor product is given by $x \cdot (u \otimes v) := (x \cdot u) \otimes v + u \otimes (x \cdot v)$. Which, given distinct indices i, j, k, l , one has the following result on the action of the bilinear form on the basis elements:

$$\begin{aligned} B(x \cdot (e_i \wedge e_j), e_i \wedge e_j) + B(e_i \wedge e_j, x \cdot (e_i \wedge e_j)) &= B((x \cdot e_i) \wedge e_j, e_i \wedge e_j) + B(e_i \wedge (x \cdot e_j), e_i \wedge e_j) \\ &\quad + B(e_i \wedge e_j, (x \cdot e_i) \wedge e_j) + B(e_i \wedge e_j, e_i \wedge (x \cdot e_j)) \\ &= 0 \end{aligned} \quad (5.8)$$

(Note: this is due to the fact that either e_i or e_j occurring in both entries for every elements in the sum).

$$\begin{aligned}
B(x \cdot (e_i \wedge e_j), e_i \wedge e_l) + B(e_i \wedge e_j, x \cdot (e_i \wedge e_l)) &= B((x \cdot e_i) \wedge e_j, e_i \wedge e_l) + B(e_i \wedge (x \cdot e_j), e_i \wedge e_l) \\
&\quad + B(e_i \wedge e_j, (x \cdot e_i) \wedge e_l) + B(e_i \wedge e_j, e_i \wedge (x \cdot e_l)) \\
&= B\left(\sum_{r=1}^4 a_{ri} \cdot e_r \wedge e_j, e_i \wedge e_l\right) \\
&\quad + B\left(e_i \wedge e_j, \sum_{r=1}^4 a_{ri} \cdot e_r \wedge e_l\right) \\
&= a_{ki} \cdot B(e_k \wedge e_j, e_i \wedge e_l) + a_{li} \cdot B(e_i \wedge e_j, e_k \wedge e_l) \\
&= 0
\end{aligned} \tag{5.9}$$

(Note: Since $(e_k \wedge e_j) \wedge (e_i \wedge e_l) = -(e_i \wedge e_j) \wedge (e_k \wedge e_l)$, this implies $B(e_k \wedge e_j, e_i \wedge e_l) = -B(e_i \wedge e_j, e_k \wedge e_l)$).

$$\begin{aligned}
B(x \cdot (e_i \wedge e_j), e_k \wedge e_l) + B(e_i \wedge e_j, x \cdot (e_k \wedge e_l)) &= B((x \cdot e_i) \wedge e_j, e_k \wedge e_l) + B(e_i \wedge (x \cdot e_j), e_k \wedge e_l) \\
&\quad + B(e_i \wedge e_j, (x \cdot e_k) \wedge e_l) + B(e_i \wedge e_j, e_k \wedge (x \cdot e_l)) \\
&= B\left(\sum_{r=1}^4 a_{ri} \cdot e_r \wedge e_j, e_k \wedge e_l\right) \\
&\quad + B\left(\sum_{s=1}^4 a_{sj} \cdot e_i \wedge e_s, e_k \wedge e_l\right) \\
&\quad + B\left(e_i \wedge e_j, \sum_{m=1}^4 a_{mk} \cdot e_m \wedge e_l\right) \\
&\quad + B\left(e_i \wedge e_j, \sum_{n=1}^4 a_{nl} \cdot e_k \wedge e_n\right) \\
&= a_{ii} \cdot B(e_i \wedge e_j, e_k \wedge e_l) + a_{jj} \cdot B(e_i \wedge e_j, e_k \wedge e_l) \\
&\quad + a_{kk} \cdot B(e_i \wedge e_j, e_k \wedge e_l) + a_{ll} \cdot B(e_i \wedge e_j, e_k \wedge e_l) \\
&= \pm 1 \cdot (a_{ii} + a_{jj} + a_{kk} + a_{ll}) = \pm 1 \cdot \text{tr}(x) = 0
\end{aligned} \tag{5.10}$$

(Note: This relies on the fact that $B(e_i \wedge e_j, e_k \wedge e_l) = \pm 1$ depending on the orientation, and $x \in \mathfrak{sl}(4, \mathbb{C})$, so $\text{tr}(x) = 0$).

Then, the above three cases showed that B is invariant under the action of any element $x \in \mathfrak{sl}(4, \mathbb{C})$ on W .

- (4) Given the Lie-algebra representation $\rho : \mathfrak{sl}(4, \mathbb{C}) \rightarrow \mathfrak{gl}(\mathbb{C}^4 \otimes \mathbb{C}^4)$, this descends to a Lie-algebra representation $\bar{\rho} : \mathfrak{sl}(4, \mathbb{C}) \rightarrow \mathfrak{gl}(\Lambda^2 \mathbb{C}^4)$. Which, the result in part (3) shows that the bilinear form B is invariant under the $\mathfrak{sl}(4, \mathbb{C})$ -action, hence any $x \in \mathfrak{sl}(4, \mathbb{C})$ satisfies $\bar{\rho}(x) \in \mathfrak{g}$ (since $\mathfrak{g}$ collects all operators on W such that the bilinear form B is invariant under it). So, $\bar{\rho} : \mathfrak{sl}(4, \mathbb{C}) \rightarrow \mathfrak{g} \cong \mathfrak{so}(6, \mathbb{C})$ is a Lie-algebra homomorphism.

Which, since $\dim \mathfrak{sl}(4, \mathbb{C}) = 4^2 - 1 = 15 = \frac{6 \cdot 5}{2} = \dim \mathfrak{so}(6, \mathbb{C})$, it suffices to check if such morphism is injective or not.

Given any $x = (a_{ij}) \in \mathfrak{sl}(4, \mathbb{C})$ (where the matrix is with respect to the standard basis), one can see their actions on the standard basis of $W = \Lambda^2 \mathbb{C}^4$ are given as follow (for simplicity here only shows for $e_1 \wedge e_2$):

$$\begin{aligned}
\bar{\rho}(x) \cdot (e_1 \wedge e_2) &= (x \cdot e_1) \wedge e_2 + e_1 \wedge (x \cdot e_2) = \sum_{i=1}^4 a_{i1} e_i \wedge e_2 + \sum_{j=1}^4 a_{j2} e_1 \wedge e_j \\
&= (a_{11} + a_{33}) e_1 \wedge e_2 + a_{31} e_3 \wedge e_2 + a_{41} e_4 \wedge e_2 \\
&\quad + a_{32} e_1 \wedge e_3 + a_{42} e_1 \wedge e_4
\end{aligned} \tag{5.11}$$

Which, if $\bar{\rho}(x) = 0$, the above two terms is 0. Then, by linear independence of the basis elements, one has $a_{31} = a_{41} = a_{32} = a_{42} = 0$, while $a_{11} + a_{33} = 0$.

Run the computation through all the basis elements, one gets that $a_{ij} = 0$ for all $i \neq j$, while $a_{ii} + a_{jj} = 0$ for all i, j . Then, given any i, j, k , since $a_{ii} + a_{jj} = a_{ii} + a_{kk} = a_{jj} + a_{kk} = 0$, one has $a_{ii} - a_{jj} = 0 = -a_{ii} - a_{jj}$, so $2a_{ii} = 0$, or $a_{ii} = 0$.

Hence, with all entries being 0, $x = 0$, showing the homomorphism $\bar{\rho}$ is injective, hence an isomorphism.